Power BI Session 2 - Student Exercises

Introduction to DAX — Measures — Calculated Columns — Advanced Analytics

Dataset: Business_Sales_Dataset_500rows.csv
Duration: 2-3 hours — **Level:** Intermediate

EXERCISE 1: UNDERSTANDING MEASURES VS CALCU

Objective: Understand the difference between measures and calculated columns in DAX.

Task:
Create both a calculated column and a measure, then analyze their differences.

Questions:

a) Create a calculated column named `Revenue_Calculation` with formula:

```
= [Quantity] * [Unit_Price]
```

Now answer:

- What does this calculated column represent?

  _____

- Does every row have a value for this column?

  _____

- What is the value for the first row? _____

b) Create a measure named `Total_Revenue_Measure` with formula:

```
= SUM([Revenue])
```

Now answer:

- What does this measure calculate?

  _____

- Can you put this measure in a table and see individual row values?

  _____

- What is the total value of this measure?

  _____

c) Compare the calculated column and measure:

- Which one evaluates for each row individually?

  _____

- Which one changes based on the current filter context?

  _____

- When would you use each one?

  _____

d) When is it better to use a calculated column instead of a measure?

_____

EXERCISE 2: CREATING BASIC MEASURES

Objective: Create fundamental measures for business metrics using aggregation functions.

Task:
Create the following measures using basic DAX aggregation functions:

Measure 1: Total Profit

```
= SUM([Profit])
```

Questions:

a) What is the Total Profit value?

b) Filter by Product_Category = "Electronics". What is the Total Profit now?

c) How does the measure value change when you apply a filter?

Measure 2: Average Order Value

```
= AVERAGE([Revenue])
```

Questions:

d) What is the Average Order Value across all orders?

e) Filter by Sales_Channel = "Online". What is the AOV for Online?

f) Filter by Sales_Channel = "Retail". What is the AOV for Retail?

g) Which sales channel has a higher average order value?

Measure 3: Total Order Count

```
= COUNTROWS(Sales)
```

Questions:

h) How many total orders are in your dataset?

i) Filter by Region = "North". How many orders in the North region?

j) What percentage of total orders come from the North region?

EXERCISE 3: CALCULATED COLUMNS WITH DAX

Objective: Create calculated columns to derive new insights from existing data.

Task:
Create the following calculated columns:

Column 1: Profit Margin Percentage

```
= DIVIDE([Profit], [Revenue], 0) * 100
```

Questions:

a) What is the profit margin for the first order?

56.94

b) What is the profit margin for the order with highest revenue?

59.76

c) Find an order with negative profit margin. What does this indicate?

-189.18 decreasing profit

d) What is the average profit margin across all orders?

50.03

Column 2: Revenue Category

```
= IF([Revenue] >= 5000, "High",
   IF([Revenue] >= 2000, "Medium", "Low"))
```

Questions:

e) How many orders are in each Revenue Category? (Count for each)

- High: 285
- Medium: 166
- Low: 49

f) What percentage of orders are in the "High" revenue category?

57%

g) Does the High revenue category represent high profit as well? Explain.

no because medium revenue is higher than it (170)

Column 3: Discount Applied Flag

```
= IF([Discount_Percent] > 0, "Yes", "No")
```

Questions:

h) How many orders have discounts applied?

396

---

i) What percentage of orders include a discount?

79.2 %

---

j) Do discounted orders have lower profit margins? Create a simple analysis.

no , discount orders have higher profit margin

EXERCISE 4: INTERMEDIATE MEASURES - FILTER CO

Objective: Create measures that are aware of filter context using CALCULATE.

Task:
Create measures that respond to filters in different ways:

Measure 1: Revenue for Selected Product Category

```
= CALCULATE(SUM([Revenue]))
```

RFCP_one = CALCULATE(SUM('Business dataset'[Revenue]),'Busir
dataset'[Product_Category] ="Electronics")

Measure 2: Revenue for All Categories (Ignoring Filter)

```
= CALCULATE(SUM([Revenue]), ALL(Product_Category))
```

Measure 3: Percentage of Total Revenue

```
= DIVIDE(
    CALCULATE(SUM([Revenue])),
    CALCULATE(SUM([Revenue]), ALL(Product_Category)),
    0
) * 100
```

Questions:

a) Filter by Product_Category = "Electronics":

- Revenue for Selected Category: _____
- Revenue for All Categories: _____
- Percentage of Total: _____

b) Filter by Product_Category = "Clothing":

- Revenue for Selected Category: _____
- Revenue for All Categories: _____
- Percentage of Total: _____

c) What is the difference between Measure 1 and Measure 2?

_____

d) When would you use the "Percentage of Total" measure?

_____

EXERCISE 5: CONDITIONAL MEASURES

Objective: Create measures that apply business logic using IF and SWITCH functions.

Task:
Create conditional measures that evaluate based on business rules:

Measure 1: High Profit Orders Count

```
= COUNTX(
    FILTER(Sales, [Profit] > 1000),
    1
)
```

Questions:

a) How many orders have profit greater than $1000?

_____

b) What percentage of total orders is this?

_____

c) Filter by Region = "West". How many high profit orders in this region?

_____

Measure 2: Profit Status

```
= IF(
    [Total Profit] > 0,
    "Profitable",
    IF(
        [Total Profit] = 0,
        "Break Even",
        "Loss"
    )
)
```

Questions:

d) What is the overall Profit Status? _____

e) Filter by Product_Category. Is every category profitable?

_____

f) Which product category has the worst profit status?

_____

Measure 3: Sales Performance Rating

```
= SWITCH(
    TRUE(),
    [Total Revenue] >= 500000, "Excellent",
    [Total Revenue] >= 300000, "Good",
    [Total Revenue] >= 100000, "Fair",
    "Poor"
)
```

Questions:

g) What is the overall Sales Performance Rating? _____

h) Filter by Region and find the rating for each region:

- North: _____
- South: _____
- East: _____
- West: _____
- Central: _____

i) Which regions are "Excellent" performers?

_____

EXERCISE 6: TIME INTELLIGENCE MEASURES

Objective: Create measures that analyze data over time using time intelligence functions.

Task:
Create measures that perform time-based analysis:

Measure 1: Year-to-Date Revenue

```
= TOTALYTD(SUM([Revenue]), [Date])
```

Questions:

a) What is the Year-to-Date (YTD) revenue for 2024?

545,952.20

b) Is this value larger or smaller than a monthly revenue value? Why?

It is larger because it includes many months, not just one.

Measure 2: Revenue Same Period Last Year

```
= CALCULATE(
    SUM([Revenue]),
    SAMEPERIODLASTYEAR([Date])
)
```

Questions:

c) What was the revenue in the same period last year?

192,926.36

d) Did revenue increase or decrease year-over-year?

the revenu dicrease for sure

Measure 3: Year-over-Year Growth

```
= DIVIDE(
    CALCULATE(SUM([Revenue])) -
        CALCULATE(SUM([Revenue]),
            SAMEPERIODLASTYEAR([Date])),
```

```
    CALCULATE(SUM([Revenue]),
        SAMEPERIODLASTYEAR([Date])),
    0
) * 100
```

Questions:

e) What is the Year-over-Year growth percentage?

95.66%

---

f) Filter by Product_Category = "Electronics". What is the YoY growth?

84.65%

---

g) Which product category has the highest YoY growth?

home and garden

EXERCISE 7: COMPLEX DAX - PROFIT ANALYSIS

Objective: Create sophisticated measures combining multiple DAX concepts.

Task:
Create advanced measures for profit analysis:

Measure 1: Average Profit per Order

```
= DIVIDE(SUM([Profit]), COUNTROWS(Sales), 0)
```

Measure 2: Profit Margin Percentage

```
= DIVIDE(SUM([Profit]), SUM([Revenue]), 0) * 100
```

Measure 3: Orders Above Average Profit

```
= COUNTX(
    FILTER(Sales, [Profit] > [Average Profit per Order]),
    1
)
```

Questions:

a) What is the Average Profit per Order?

_____

b) What is the overall Profit Margin Percentage?

_____

c) How many orders are above the average profit threshold?

_____

d) Filter by Sales_Channel = "Online". Compare these metrics to the overall:

- Online Avg Profit per Order: _____
- Online Profit Margin: _____
- Orders Above Avg Profit: _____

e) Which sales channel is more profitable - Online or Retail?

_____

EXERCISE 8: MEASURE EVALUATION DEBUGGING

Objective: Understand how measures are evaluated and troubleshoot common issues.

Task:
Analyze the behavior of measures in different contexts:

a) Create a table with Region and Revenue (measure). What do you observe?

- Does each region show different revenue values?

_____

- Why doesn't the measure show the same value for each row?

_____

b) Create a table with Region and a calculated column "[Quantity] * [Unit_Price]". What do you observe?

- Is every row different?

_____

- Why does the calculated column show different values for each row?

_____

c) Create a measure with the formula: `= SUM([Quantity])`

- Put this measure in a card. What value does it show?

_____

- Filter by Region = "North". What value does it show now?

_____

- Explain how the filter changed the measure result:

_____

d) Try this measure: `= SUM([Quantity]) / COUNTROWS(Sales)`

- What does this measure calculate? (Hint: Average Quantity per Order)

  _____

- Does the result change when you filter by Region?

  _____

EXERCISE 9: REAL-WORLD BUSINESS MEASURES

Objective: Create practical business measures used in real dashboards.

Task:
Create the following business intelligence measures:

a) **Discount Impact on Profit**

Create a measure that calculates profit loss from discounts:

```
= SUM([Discount_Amount])
```

- What is the total discount given across all orders?

- Filter by Product_Category = "Clothing". How much discount was given?

- If the company eliminated all discounts, how much additional profit would they have?

b) **Shipping Efficiency**

Create a measure for average shipping cost per order:

```
= AVERAGE([Shipping_Cost])
```

- What is the average shipping cost?

- Which region has the highest average shipping cost?

- Why might some regions have higher shipping costs?

c) **Customer Segment Performance**

Create a calculated column that segments orders by profit:

```
= IF([Profit] > 2000, "High Value",
   IF([Profit] > 500, "Medium Value", "Low Value"))
```

- How many orders in each segment?
  - High Value: _____
  - Medium Value: _____
  - Low Value: _____
- What percentage of orders are "High Value"?

  _____

- Which sales channel produces the most high-value orders?

  _____

EXERCISE 10: COMPREHENSIVE DAX PROJECT

Objective: Apply all DAX concepts to build a comprehensive analytics measure set.

Task:
Create an Executive Dashboard with these measures:

a) Create 5 KPI measures and record their values:

   i. Total Revenue: _____
   ii. Total Profit: _____
   iii. Profit Margin %: _____
   iv. Avg Order Value: _____
   v. Total Orders: _____

b) Create 3 analytical measures for performance analysis:

   i. Orders Above Average Profit
   ii. Percentage of Total Revenue (by category when filtered)
   iii. Average Profit per Region

   Record values:

   - Orders Above Average Profit: _____
   - Best Region by Avg Profit: _____
   - Worst Region by Avg Profit: _____

c) Create a dashboard with:

   - Top row: 5 KPI cards
   - Middle: Revenue by Product Category (with measure)
   - Bottom: Profit by Sales Channel (with measure)
   - Add slicers: Region, Product Category, Year

d) Analyze the dashboard and answer:

   - Which product-region-channel combination is most profitable?

     _____

   - What is the most profitable sales channel overall?

     _____

- Based on your measures and analysis, what is ONE business recommendation?

REFLECTION QUESTIONS

Answer these to demonstrate your DAX understanding:

1. **Filter Context vs Row Context:**
   - What is filter context? Give an example from your exercises.

     _____

   - What is row context? Give an example from your exercises.

     _____

   - How are they different?

     _____

2. **Measures vs Calculated Columns:**
   - When did you choose to use a measure instead of a calculated column?

     _____

   - What are the performance implications?

     _____

3. **DAX Functions Used:**
   - List 5 DAX functions you used and explain what each does:
     i. _____: _____
     ii. _____: _____
     iii. _____: _____
     iv. _____: _____
     v. _____: _____

4. **CALCULATE Function:**
   - Why is CALCULATE important in DAX?

     _____

- How did you use CALCULATE in your measures?

5. **Business Impact:**
   - How do DAX measures help answer business questions?
   - What business insights did your measures reveal?

CHALLENGE EXERCISES (OPTIONAL)

For advanced learners:

1. **Nested IF Statement Challenge**

   Create a measure that classifies regions into performance tiers:

```
= SWITCH(TRUE(),
    SUM([Revenue]) >= 300000, "Tier 1 - Elite",
    SUM([Revenue]) >= 200000, "Tier 2 - Strong",
    SUM([Revenue]) >= 100000, "Tier 3 - Growing",
    "Tier 4 - Emerging"
)
```

   What tier does each region belong to?

---

2. **Complex CALCULATE with Multiple Filters**

   Create a measure comparing current selection to a specific region (North):

```
= DIVIDE(
    SUM([Revenue]),
    CALCULATE(SUM([Revenue]), Region = "North"),
    0
)
```

   How does each region compare to North?

---

3. **RANKX Function Challenge**

   Create a measure that ranks regions by profit:

```
= RANK([Total Profit], Table_Name)
```

   Rank all regions by profit (1 = highest, 5 = lowest).

4. **DISTINCTCOUNT Challenge**

   Create a measure counting unique customers:

```
= DISTINCTCOUNT([Customer_Name])
```

   How many unique customers purchased in each region?

---

5. **Year-over-Year Growth Analysis**

   Create measures comparing 2023 vs 2024 revenue with percentage growth.

   - 2023 Revenue: _____
   - 2024 Revenue: _____
   - YoY Growth %: _____

SUBMISSION CHECKLIST

Before submitting your work, verify you have completed:

Exercise 1: Measures vs Calculated Columns understood with all questions answered

Exercise 2: Basic measures (Total Profit, Average Order Value, Order Count) created

Exercise 3: Calculated columns (Profit Margin, Revenue Category, Discount Flag) with Q&A

Exercise 4: Filter context measures (CALCULATE) with comprehensive analysis

Exercise 5: Conditional measures (IF, SWITCH) with business analysis

Exercise 6: Time intelligence measures (YTD, Year-over-Year) with historical analysis

Exercise 7: Complex measures combining multiple DAX concepts

Exercise 8: Measure evaluation and debugging analysis completed

Exercise 9: Real-world business measures with actionable insights

Exercise 10: Comprehensive DAX project with executive dashboard

All Reflection Questions answered thoroughly

Report is organized, clean, and demonstrates DAX mastery

Congratulations!
You've mastered DAX fundamentals!